

Unit 3 : Concept Development and Design Methodology

Product information gathering, Brainstorming and lateral thinking, Concept generation and selection, Design evaluation and technical feasibility estimation, Pugh's concept selection charts and concept scoring, Process of concept embodiment, Simulation driven design, Fast method and subtract-operate procedure.

Product Information Gathering

This activity is necessary for dynamic search for data that will contribute to the technology, physical principles, and industrial designs of a product.

A classification of informational sources can be classified as below:

- i) Benchmarking
- ii) Analogies-a) Nature b) Product function c) Product architecture
- iii) Published media-a) Patent b) Journals c) Product information d) Textbooks
- iv) People-a) Customers b) Experts c) Professionals in the field
- v) World Wide Web (www)- a) Manufacturing Listening's b) Information research services

A patent search is essential in understanding the current technology. It is also useful for seeking analogous ideas from products that solve similar functions.

Brainstorming and Lateral Thinking

- Brainstorming is an intuitive method for generating concepts.
- It focuses on product function and architecture, where team members communicate verbally during a set time period.
- All team members are open encouraged to be open during the initial session of brainstorming.
- There is no need to strictly adhere to product specifications but instead of this, it is required to focus on the functional needs of the product.

Guidelines that should be followed during a brainstorming session:

- i) This is Designate a group leader/facilitator to prevent judgments and encourage participation by all. The facilitator should not contribute directly but record all the discussion during the session.
- ii) Form the groups with 5-15 people, usually no more or no less in members and should not give inadequate ideas.
- iii) Brainstorming session should be of 30-45 minutes. First 10 minutes should be typically devoted to problem orientation and familiarity. Next 20-25 minutes will see a sharp increase of ideas.
- iv) Don't confine the group to experts in the new area. The only way to obtain new ideas is to introduce new knowledge.
- v) Depending on the goals of the brainstorming session, individuals may enter a session, will a set of ideas.

vi) Avoid hierarchy structured groups. Bosses, supervisors and managers should not be induced in many sessions.

Steps for brainstorming:

- i) Announce the topic, problem.
- ii) Take a few minutes to think and write.
- iii) Everyone contribute ideas.
- iv) Any additional suggestions are added.
- v) Discuss ideas to narrow choices.

Lateral Thinking

- It's a term first used by Edward de Bono in 1967. It defines lateral thinking as a method of thinking concerned with changing concepts and perception.
- Edward de Bono defines four types of thinking:
 - **Idea Generating Tools:** Intended to break current thinking pattern routines.
 - **Focus tools:** Intended to broaden where to search for new ideas.
 - **Tools:** Intended to ensure more value is received from idea-generating output.
 - **Tools:** That promote consideration of real-world constraints, resources, and support.
- **Random Entry Idea Generating Tools:** A thinker chooses an object at random and associates it with the area they are thinking about.
- **Provocation Idea Generating Tools:** The provocation includes wishful thinking, exaggeration, reversal, escape, or distortion. The thinker creates a list of provocations and then uses the most outstanding to move their thinking forward to new ideas.

Morphological Analysis

- A system method for assembly and continued idea generation is known as morphological analysis.
- Morphological analysis is a tool that provides structured search and combination of concepts in product design.
- The process of executing this analysis is as below:
 - i) Consider each product function in a functional model and each model of product architecture.
 - ii) List the function or module as a row of a matrix.
 - iii) In the first column of the matrix, enter the current solution to the function, if the product exists.
 - iv) Apply a concept-generation method and record the concepts in the column of the matrix, for each function.
 - v) Map the range of solutions per each function to a classification scheme. Judge if the solutions are too focused or cover a good

breadth, carry out further sessions of intuitive and direct concept generation.

vi) When a good breadth of ideas and technologies are realized in the morphological matrix, combine the ideas into diverse concept variants that seek to satisfy the entire product specification.

Seven steps of morphological analysis

There are seven steps in Morphological analysis as follows:

- **Step 1: Feasibility study**

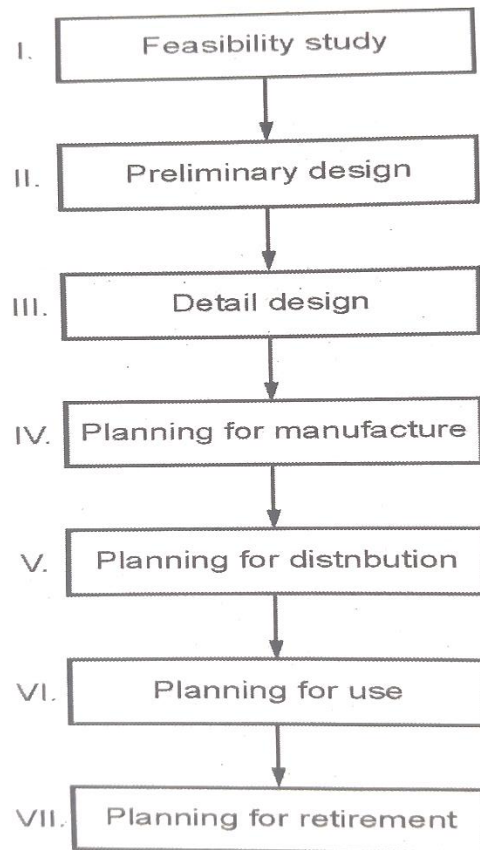
- This stage is also called conceptual design. It always begins with a feasibility study.
- The purpose and activities during a feasibility study are:
 - To ascertain if there really exists a need (i.e. the existence of a need must be supported by necessary pieces of evidence, rather than the outcome of one's imagination).
 - Search for a number of possible solutions.
 - Evaluate the solutions i.e., is it physically realizable?
 - Is it economically worthwhile?
 - Is it within our financial capacity?

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- **Step 2: Preliminary (Embodiment) design**

- This is the stage at which the concept generated in the feasibility study is carefully developed.
- The important actions done at this stage are:
 - Model building and testing.
 - Study the advantages and disadvantages of different solutions.
 - Check for performance, quality, strength, aesthetics, etc.

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- **Step III: Detail design**

- Its purpose is to furnish the complete engineering description of the tested product. The arrangement from, dimensions, tolerances and surface properties of all individual parts are determined. Also, the materials to be used and the manufacturing process to be adopted are decided. Finally, a complete prototype is tested.

- **Step IV: Planning for manufacture**

- This phase includes all the production planning and control activities necessary for the manufacture of the product.

The main tasks at this phase are:

- Preparation of a process sheet i.e., the document containing a sequential list of manufacturing processes.
- Specify the condition of raw materials.
- Specify tools and machine requirements.
- Estimation of production cost.
- Specify the requirement in the plant.
- Planning QC systems.
- Planning for production control.
- Planning for information flow system etc.

Step V: Planning for distribution

The economic success of a design depends on the skill exercised in marketing. Hence, this phase aims at planning an effective distribution system. Different activities of this phase are:

- Designing the packing of the product.
- Planning effective and economic warehousing systems.
- Planning advertisement techniques.
- Designing the product for effective distribution in the prevailing conditions.

Step VI: Planning for consumption/use

The purpose of this phase is to incorporate into the design all necessary user-oriented features. The various steps are:

- Design for maintenance.
- Design for reliability.
- Design for convenience in use.
- Design for aesthetic features.
- Design for prolonged life.
- Design for product improvement on the basis of service data.

Step VII: Planning for retirement

This is the phase that takes into account when the product has reached the end of its useful life. A product may retire when:

- It does not function properly.
- Another competitive design emerges.
- There are changes in taste or fashion

Generating Concepts

Firstly

- The function structure and product architecture of existing or original products are created.
- A morphological matrix is created and labeled with each row with product sub-function.
- Immediately after this, the first column of the matrix is filled with current solutions (if they exist).
- For each product function (each row), consider the existing product and identify the component that satisfies the function.
- After forming the initial matrix, blackout functions that are not part of the redesign effort. This action is usually carried out for three reasons:
 - a) The product functions solution component cannot be changed.
 - b) The product functions solution components are not part of the redesign effort and are not conceptual, customer, or too expensive to invest resources.
 - c) Support functions might not be considered at this stage in the development process.

Design Evaluation:

- Over the course of the product development process, many persons make many decisions.
- Many of these decisions can be simply made by a single engineer or a small group of engineering based on information that is readily accessible.
- The decisions follow from earlier decisions and from the experience of the engineers all in logical progression.
- On the other hand, there are important decisions made by engineers where it is not apparent what the decisive outcome should be.
- There are many alternative possibilities and many criteria that the decision impacts. Such difficult decisions are called as design evaluation.
- Design evaluation occurs at all phases of the development process, from the concept phase through the detailed design phase.
- Design evaluation cannot be effectively made without a structured decision-making process.

Estimation of Technical Feasibility

- After a complex analysis is completed, or during the preliminary concept evaluations, the ability to compare the results with known quantities and judge the decision's validity is critical.
- There are four basic steps to estimate, imagine, model, compare, and judge.
- The first step is to imagine the concept to estimate. In this step, a picture is created of what happens with the concepts along with energy, material, or information flows through the concept, where you can measure an input/output change.
- The second step is to construct a very simple model that relates the capacity or flow (energy, material or information) through a concept to known quantities.
- The third step is to use the concept and model to provide a comparison with known quantities.
- In the model, imagined values are filled in on the model variables, and the comparison value is calculated in the dimensions of the known quantities.

Solution	1	2	3	4	5	n
Sub Section						
1						
2						
3						
4						
m						

General Morphological Matrix Format

Concept Selection Process

- Generally concept selection is a critical point in product development process.
- There is no clear solution concept that should be perceived, different members of team have strong opinion on different solution.
- The selection process is 5 step process.

Step 1 : Forming consensus on the criteria

- The first step is to establish, evaluation criteria on which the concept selection decision will be based.
- Most concept selection decision are based on evaluation criteria of the cost.
- Then development risk, technical difficulty or ability to meet schedule delivery and finally performance or customer satisfaction.
- The problem is that once, the criteria are stated, every design team member will have different perception of what these statement mean.
- An initial task is to resolve, there differences into definitions that are common to the team.
- To establish criteria definitions, a design team should start with one member articulating a proposed list of evaluation criteria.
- This list should be developed from the customer needs and engineering specifications of products.
- It should also includes most critical and important specifications.

Step 2 : Farming Consensus on the alternatives

- Once the evaluation criteria is initially established, different alternatives need to be understood on a common basis.
- First team members voice alternatives that are given labels where only a rough understanding is required by the members.
- These alternatives should be those that proceed from concept generation.
- They should also be refined to a level where all team members may understand them, without any difficulty.

Step 3 : Ranking

- The next step is to rank each clearly defined alternatives on each clearly defined criteria.
- The rankings are completed using a decision matrix.
- A team should consider each criterion one at a time and rank all the different alternatives on each criteria.
- A team should not rank each alternative across all criteria before considering another alternative.
- A scale used such as (-ve) is worst and (+ve) is good.

Step 4 : Assessment

- Once rankings are completed for each criterion, the evaluation should be collected into overall summary ranking on each alternatives.
- The basic reason for creating summaries is that human can not process all the ranking simultaneously when comparing two alternatives.
- The criterion ranking need to be aggregated into one or more overall ranks for simpler ordering of alternatives into a best to course ranking that can be understood.
- Generally these aggregation is a simple exercise in the mathematics of ranking to see how the alternative rate overall.

Step 5: Attaching the negatives

- Once the alternatives are ordered by rank, the process is not finished; rather the alternatives that rate poorly should be taken off and other alternatives that rate favourably must be more closely examined.
- In short, an alternative that rate high overall but have a few low scores should be closely scrutinized.
- Pugh (1990) used this process attaching the negatives. It is critical and should not be minimized is importance.
- For the alternatives with such negative ranked criteria, a design team should clearly state what is causing the negative ranking as a chain of physical effects.
- Than each of these effects should be investigated by the team to understand how each engineer might apply their discipline to eliminate the negatives.

Pugh Concept - Selection Charts

- Pugh charts are more effective known tools for preliminary concept selection when there is a minimum information quantity available.
- They are also effective as the information quality increases and the selection scale is refined.
- In Pugh chart concept sketches are displayed at the top of the matrix as columns.
- Each criterion is list down in each row.

Criteria	Concepts					
	Criterion 1	Sketch of Datum	Sketch of 'B'	Sketch of 'C'	Sketch of 'D'	Sketch of 'E'
	Criterion 2	0				
	Criterion 3	0				

	Criterion 4	0				
	Criterion 5	0				
	$\Sigma + (P_i)$	0 +				
	$\Sigma - (N_i)$	0				

Template of pugh selection chart

i) Select a datum: The next step is to establish the evaluation scale.

- An effective process is to construct simple evaluation scale.
- To do this team should select one alternative that will be ranked as (s) or (o) on energy criteria and be called datum.
- The datum is an alternative to which every other concept will be compared.
- The other concepts are rated as either worse (-ve), the same (s) or better on each criterion.
- To select a datum, a team can choose one of several approach: a) A competitive product that team wishes might serve as a datum. b) The current product offered by the company might serve as a well understood concept. c) Pugh recommends using the alternative that the team votes as best as the datum.

ii) Ranking and Assessment: Having selected a datum, all the alternatives are evaluated on each criteria, one criterion at a time relative to the datum.

- If team agrees that the alternatives offer worst performance than the datum then the alternative are assigned negative rank (-ve).
- If team agrees that the alternatives offer better performance than the datum then assigned as positive (+ve) rank. If performance of alternatives offers the same as datum, it is assigned (s) rank.

iii) Alternative Rank Ordering: Each criterion is not equally important.

- A (-ve) on one alternative may really be (... -ve) and a (+) on another hand may be only one half of a (+) compared with each other.
- To deal with this discrepancy, one should use three overall scores to compare the alternatives on each alternative (i) one can determine the average, overall score.
 - $S_i = \Sigma R_{ij}$
- In addition to more overall scores should be calculated, where each adds up all negative (-ve) scores and all positive (+ve) scores.
 - $P_i = \Sigma R_{Pj}$

$$N_i = \sum_{j=1}^J R_{ij}$$

iv) Attacking the negatives: The lower average rated alternatives should be discarded from further consideration.

- Then the highly average rated alternative with (-ve) scores as reflected by 'N_i' scores should be scrutinized with hope of out of box thinking for new alternatives or combinations, that eliminates the negative ranks.

v) Iteration and solution:

- The team converges to consensus on preferred alternatives, were energy one understands the alternatives why each was rejected and accepts the judgement.
- The team comes to understand a refined choices.
- Here the team could not make the final decision because specific information is lacking. Tap or more conditional alternatives can not be distinguished because their rank on criterion can not be determining at this time.
- The team comes to understand that they do not understand the criterion, here the team can not reduce, the choice to unclear alternatives that need refining because they do not agree on criteria.

Concept Scoring

The next step after concept screening or selection is the concept scoring. This step would specify the ideas to get the best idea. In concept scoring every selection criteria in this step is would have its weight factor. The weight criteria is based on how important the criteria. The energy concept will be given rating from 1 upto 5. This rating will be multiple by weight to get weighted score. Then it require to sum up weighted score from each criterion for each concept to get total score for each concept. A concept that get higher score will be chosen to be developed.

Process of Concept Embodiment

- Embodiment design deals with specific parameters and layout that is layout design of the product.
- Embodiment design provides from to selected concepts.
- Ideally the result of concept development is a single concept with one chosen alternatives for each sub - systems. In reality usually ends up with one of three situations:
 - i) A single concept with some function choices that are not set because they are not primary.
 - ii) A family of products with single choice for each member of the family.

iii) Two or three alternative concepts might need further refinement before a choice is made.

- Embodiment design moves the process towards a definitive form including
 - i) Geometric layout
 - ii) Material composition
 - iii) Quantity and manufacturability issues
 - iv) Economics
- The process begins by considering the product specifications.
- Using customers needs and these specifications, the critical needs are identified that will drive the embodiment product.
- After choosing the specifications layout of product is drawn to scale based on the concept drawing.
- At this stage the drawing should not be fully detailed and care should be taken not to over construct the layout.
- Through these drawings following items are illustrated, maximum dimension of product clearance between relative subsystem, general arrangements of components, relative to one another.
- Once rough scale sketches are completed, the main function carries of functional model.
- It is important to check for possible simplification and combining same components.
- Based on the result preliminary alternative layout of remaining main functions are developed.
- Ranges of variables (geometry, materials, etc. should then be listed for each of the functions.
- Each subsequently should also be evaluated to determine, whether standard parts can be used for sub-assemblies or if custom parts will be needed.
- Using functional model and product architecture layout rough layouts should be developed for remaining functions.
- With rough layouts developed for the main, supporting functions, these layouts must be detailed, ensuring the compatibility of all sub-assembly.
- This tasks usually calls for the use of standard, mathematical models, design guidelines and experimentation to determine all appropriate parameter.

System Modeling

- Functional models represents high level system model of product.
- Inputs of materials, energy and signals are converted into desired outputs.
- During the concept deployment phase of a product these models provides the basis for systematically generating products architectures and solution concepts.
- These models need to be extended to facilitate design parameters and manufacturing decisions during concept embodiment.
- Mathematical methods and physical prototypes are the means to perform this.
- Virtual and physical modelling of product provide in depth insights into its operation and possible improvements.
 - i) High Level Modeling: After identifying the physical principles and assumptions for each customers need balance relationship is created to document a high level physical model.
- Cause and affinity diagram may be used to document the balance relationship.
- Where the effect is the customer need and the causes are physical principles.

B) Balance Relationship

The last task in formulating mathematical models for product is to convert the balance relationship into a set of mathematical equation.

C) Physical Prototypes Models

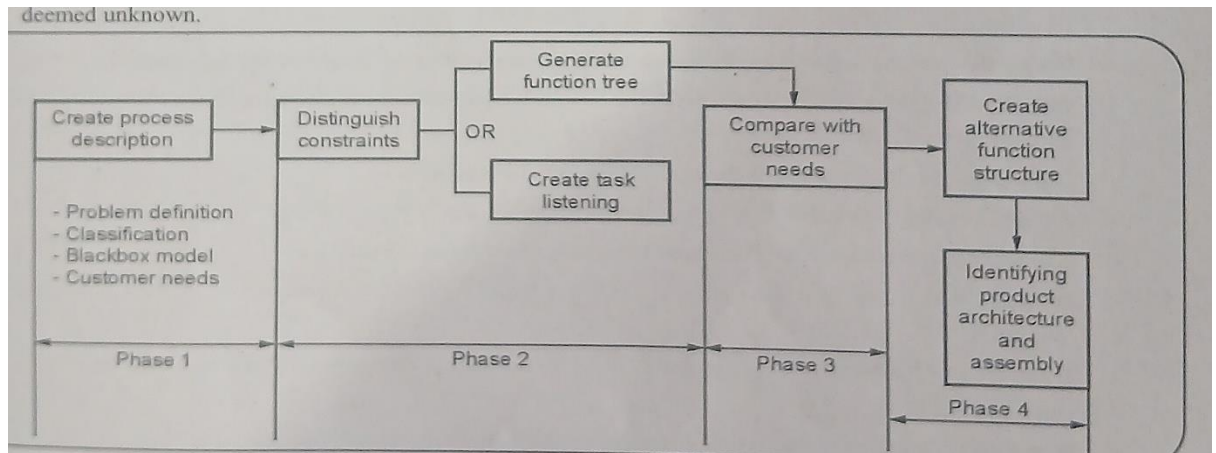
The creation of physical prototype can be used as an alternatives modelling approach.

- The intention is to create a prototype is to experiment for customer needs, focusing on the affected product components and variables.

Functional Modelling and Decomposition

- All products do have certain functionality. There is some intended reason behind their existence, i.e. the product function.
- Product do have secondary functions and constraints, such as be inexpensive but these constraints are not product functions.
- **Function** : A function of a product is a statement of clear, reproducible relationship between the available input and desired output of a product.
- The product function is the overall intended function, what is to do ?

- **Sub - function** : A sub-function is a component of product function. An overall function can and often must be divided into identifiable sub-function corresponding to subtasks.
- For all functional modelling methods a basic construct is the black box called black because it's internal form is deemed unknown.

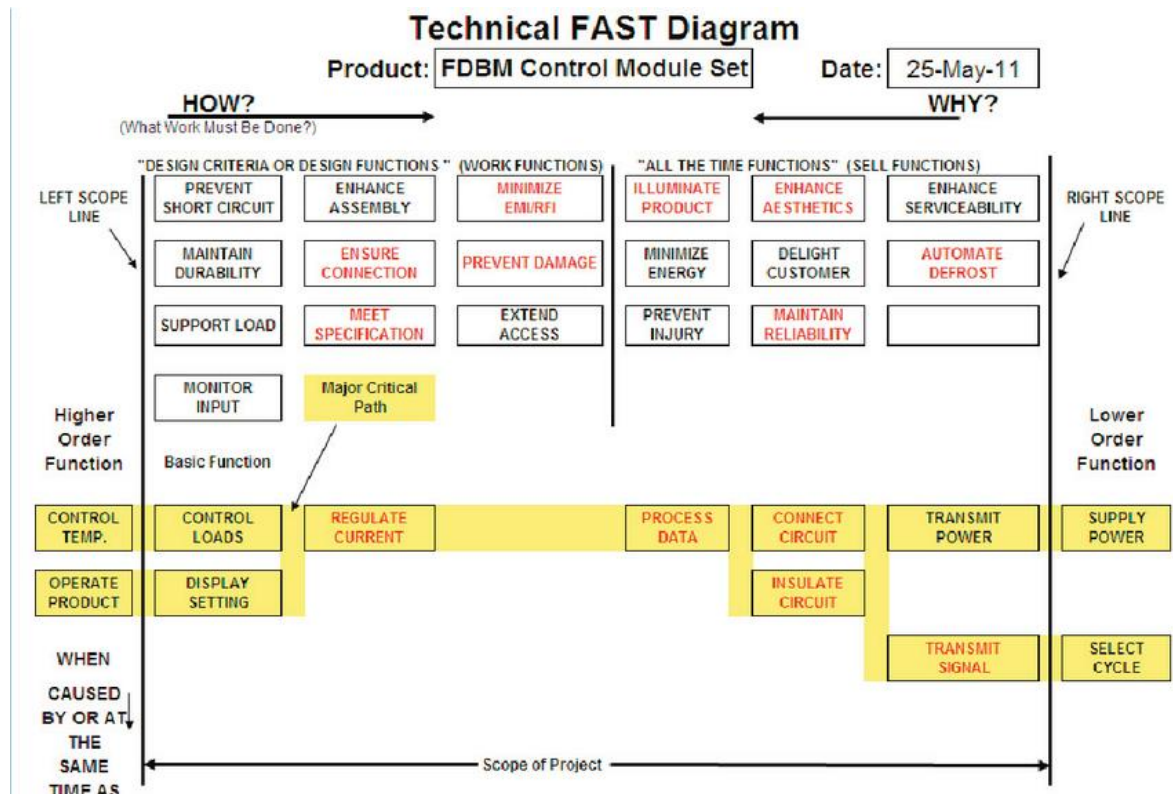


Modeling process

- One can think of the product as doing something the function by taking inputs or modifying these inputs and thereby creating outputs.
- Any black box can be expanded into many interlinked sub - boxes just as functions can be decomposed into sub - functions.
- Functional modelling as summarized in figure begins with systematic process of transforming these needs to a clear specification of design task.
- It is also initiated the conceptual design phase where in
- Notice in figure that either function (hierarchies trees) or function structure (task listening's) may be used to model a product's function.

FAST Method

- The Functional Analysis System Technique (FAST) is used to define, analyze and understand product functions, how the functions related to one another and which functions require attention to increase the product value.
- It is used to display functions in a logical sequence and prioritize them.



- During this process, it becomes obvious that these functions have different level of importance.
- Out of all functions, one function that is the overall product function has to be selected. (Sometimes called basic functions).
- The product function represents the main reason that the product exist.
- For e.g. - The basic function of a car seat belt is to restrain a person in a car seat.
- A product may have more than one base function under different conditions.
- For e.g. - An electric fuse that conducts electricity under certain conditions also break the circuit under extreme conditions.
- These creates complexity of understanding FAST incorporates this complexity by requiring separate analysis for separate conditions.
- With FAST, the function tree is established by asking "How is this function achieved."
- The answer is at at least one usually several secondary functions.
- In FAST diagram the overall function is placed at the top left of the diagram.
- The sub-functioning results from the how questioning are listed to the right of overall functions.

- The stop in modelling process (and in constructing the FAST diagram are as follows)
 - i) Construct the verticals dashed lines on both extreme left and right. These lines defines the scope of product development objectives.

ii) Place the basic function to the right of the left hand scope line, pose the question, why is the basic function being performed ? A higher order function will answer this question to the left of basic function and connect with line beginning the actual path.

iii) General functions to the right of basic function. These functions should always follow a how and why answering scheme and represent the secondary function connected these functions with lines to define further more of critical path.

iv) The critical path will end with an assumed function outside the right scope line. This function is external to the product.

v) Generate the remaining secondary functions by placing that under the functions that relate to basic functions. These functions either occur at the same time or are caused by the functions on the critical path.

vi) State the objectives of development effort above the basic function. In addition add one time or all time function to the top of diagram.

Subtract Operate Procedure

- The FAST down approach is top down approach with starts with overall function and then decomposes it.
- Rather one might wish bottom up approach than that of top down approach and subtract and operate procedure does the same.

Steps :

i) Disassemble (subtract) one component of assembly - removal of component may occur in any order. However it may be necessary to remove the desired component. These prerequisite components should be reassembled if possible. If they can not be reassembled, measure should be used to replicate the functions of missing components.

ii) Operate the system through its full range : This step should test product through the range of customer requirements. After removing a component, the product should be thoroughly tested. For each customer requirement (economical, structural) the product must be tested to verify the effects of removal of components.

iii) Analyze the Effect : This step is most commonly completed through a visual analysis. However it may necessary to use testing device.

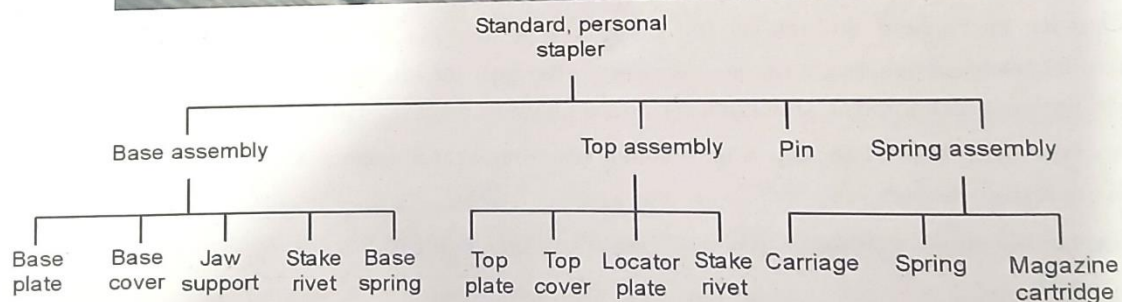
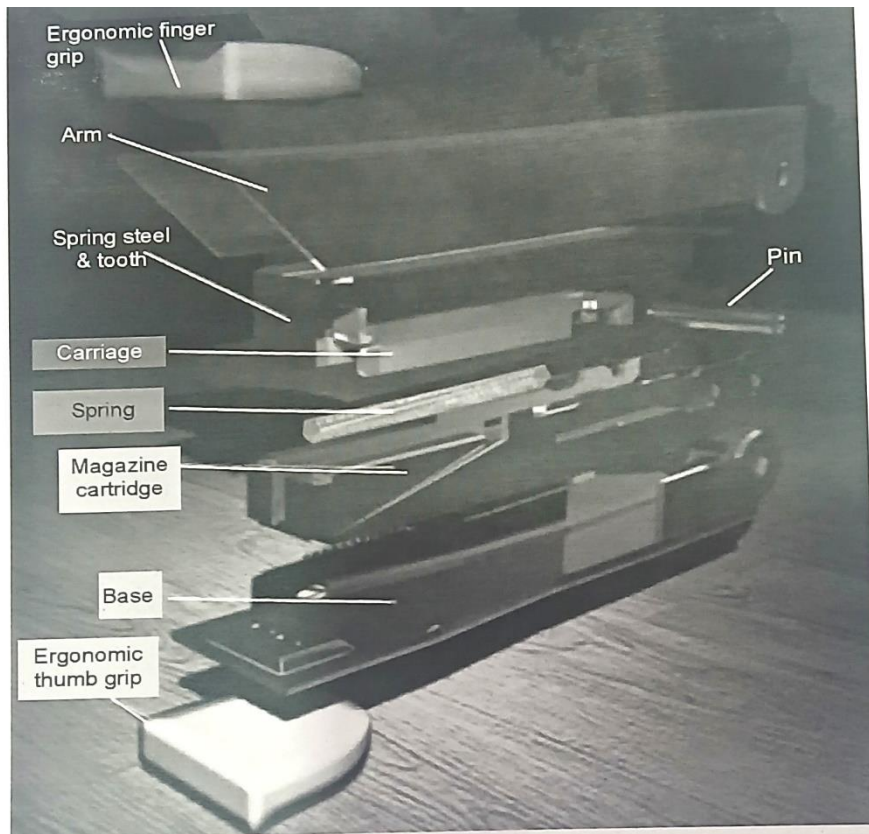
iv) Deduce the sub-function of missing component : From step (iii) the sub-function of missing component may be deduced.

v) Replace the component and repeat the procedure "ir" time, where "ir" is the number of components in assembly. Document the result in table.

vi) Translate the sub-functions into function tree : One does this by grouping the sub-functions of step (v) into common group.

Example of subtract and operate method: personal stapler

Mechanical device that joins sheets of paper by driving a thin piece of metal and folding the ends



SOP: Remove components and analyze effect

- Base assembly
- Base plate : Press-force is not supported
- Base cover : Press-force goes to surface

- Aw support : The movement is not solid
- State rivet : The staple is not formed
- Base spring : The arm does not move back

SOP: Remove components and analyze effect

- Top assembly
- Top plate : The spring assembly is loose
- Top cover : The hand grip is not comfortable
- Locator plate: The spring assembly is loose
- Stake rivet: Not enough force is created for staples
- SOP: Remove components and analyze effect
- Pin: The top and bottom assemblies are not held together
- Spring assembly
- Carriage: The cartridge does not move
- Spring: The staplers do not move forward
- Magazine cartridge: No place to put the stapler

Simulation Driven Design

- Simulation driven design is taking simulation technology and moving it from the middle and late cycle of design process to very front of it. This drastically reduces the time it takes to for companies to develop products, because of going back and forth between detailed design and validation, validation or simulation it put at the front of that process.
- The simulation is used to design the product using things like topology optimization or it is required to integrate control systems at the earliest stages and then validation is done. It's simple check box process instead of long iterative process.
- So simulation driven design is putting simulation at the front of design process and using simulations technology to create design instead of simulation to analyze it later.
- This is very much common thread in industry 4.0 enabled manufacturing and driven. In old days, it is required to wait for a design to come down stream before testing building or simulating a design. Then if the design was not upto the scratch, it is required to go back to the drawing based and try again.

- Now with the help of design and simulation software, it become possible to simulate early and decide what strategy to be used before getting too deep into the product life cycle.

Questions

1. Explain product information gathering.
2. Explain the concept of brainstorming with suitable example.
3. Explain morphological analysis with suitable example.
4. What is technical flexibility estimation? Explain.
5. Explain Iraqi concept selection charts with example.
6. Discuss the concept of functional modelling decomposition.
7. Explain FAST method, elaborate with example.
8. Explain subtract and operate procedure.
9. Explain simulation driven design.

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